

Chapter 19 :

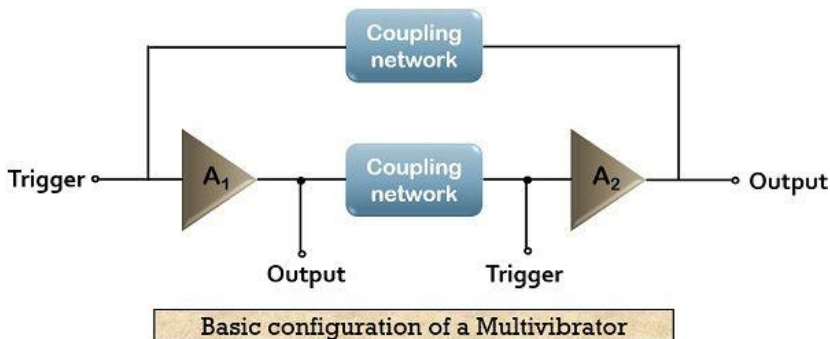
Multivibrators

19.1 Introduction

Multivibrator is one kind of oscillator that produces a non-sinusoidal waveform as its output. The generated non-sinusoidal waveforms are basically a square wave, rectangular wave, a triangular wave, sawtooth wave, or ramp wave etc. Generally, it consists of two-stage *RC* coupled amplifier that operates in two modes (ON or OFF). The modes are basically termed as **states** of the multivibrator. Multivibrators are used as frequency generators, frequency dividers and generators of time delays and also as memory elements in computers etc.

A transistor basically functions as an amplifier in its linear region. If a transistor amplifier output stage is joined with the previous amplifier stage, such a connection is said to be coupled. Thus, a multivibrator can also be defined as a two-stage *RC* coupled amplifier with positive feedback from the output of one amplifier to the input of the other. Two transistors are connected in feedback so that one controls the state of the other. Hence the ON and OFF states of the whole circuit, and the time periods for which the transistors are driven into saturation or cut off are controlled by the conditions of the circuit.

The following figure shows the block diagram of a Multivibrator.



Electronics Coach

19.2 Time Calculation of RC Circuit

In an RC circuit the capacitor charges and discharges exponentially. During the charging phase, the capacitor voltage can be calculated using the following expression.

$$v_C = \varepsilon(1 - e^{-t/RC}) \quad (19-1)$$

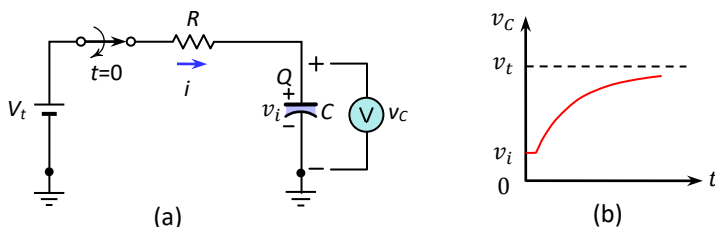


Fig.19-1: (a) Capacitor charging circuit, (b) Charging waveform

Fig.19-1(a) shows an RC circuit with the target voltage (charging voltage) of V_t . At the moment of turning the switch, the capacitor contains some charge, and due to this charge, the capacitor voltage is V_i (initial voltage). So, the effective charging voltage for the capacitor is $(V_t - V_i)$. For this situation, we can write the Equ.(19-1) as

$$v_C = V_i + (V_t - V_i)(1 - e^{-t/RC})$$

or,

$$v_C = V_i + V_t - V_i - (V_t - V_i)e^{-t/RC}$$

or,

$$v_C = V_t - (V_t - V_i)e^{-t/RC}$$

or,

$$(V_t - V_i)e^{-t/RC} = V_t - v_C$$

or,

$$e^{-t/RC} = \frac{V_t - v_C}{V_t - V_i}$$

or,

$$-\frac{t}{RC} = \ln\left(\frac{V_t - v_C}{V_t - V_i}\right)$$

or,

$$\frac{t}{RC} = \ln\left(\frac{V_t - V_i}{V_t - v_C}\right)$$

or,

$$t = RC \ln\left(\frac{V_t - V_i}{V_t - v_C}\right) \quad (19-2)$$

19.3 Astable Multivibrator

The **Astable Multivibrator** is one type of cross-coupled transistor switching circuit that has NO stable output state, as it changes from one state to another (HIGH and LOW) by its own. That

is, the output of this circuit (taken from the collector terminal of any transistor) freely oscillates between HIGH and LOW state [as shown in Fig.8-15(b)]. The astable multivibrator is also called free running multivibrator.

As shown in Fig.8-15(a), an astable multivibrator is consisted of two transistors, two capacitors and four resistors.

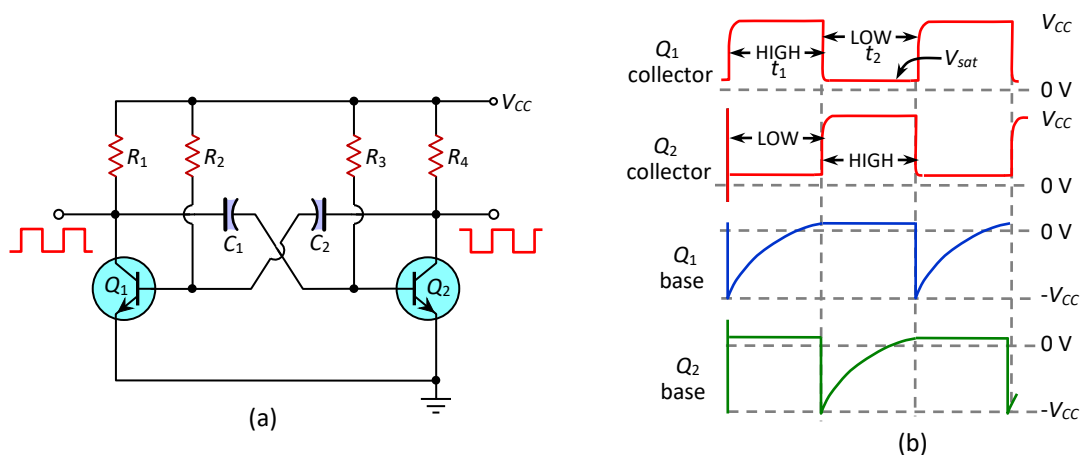


Fig.19-2: Astable multivibrator: (a) Schematic diagram, and (b) Output waveforms

The HIGH and LOW time depends on the values of R_2 , R_3 , C_1 , and C_2 . If the output is taken from the collector of Q_1 , then the high time will depend on R_2 and C_2 . This output will remain HIGH during the time C_2 charges to 0.7 V from an initial voltage of $-V_{CC}$. The capacitor will charges through R_2 resistor and the target voltage is V_{CC} . So using Equ.(19-2), we get

$$\text{or,} \quad t_H = RC \ln \left(\frac{V_t - V_i}{V_t - v_c} \right) = R_2 C_2 \ln \left(\frac{V_{CC} + V_{CC}}{V_{CC} - 0.7 \text{ V}} \right)$$

Neglecting
0.7 V

$$t_H = R_2 C_2 \ln \left(\frac{V_{CC} + V_{CC}}{V_{CC}} \right) = R_2 C_2 \ln(2)$$

$$\text{or,} \quad t_H = 0.693 R_2 C_2 \quad (19-3)$$

Similarly, the LOW time of this output will be determined by R_1 and C_1 . Using the same calculation method, we will find

$$t_L = 0.693 R_1 C_1 \quad (19-4)$$

So, the time period of this waveform will be,

$$T = t_H + t_L = 0.693 R_2 C_2 + 0.693 R_1 C_1 = 0.693 (R_1 C_1 + R_2 C_2) \quad (19-5)$$

If the values of the resistors and capacitors are assumed to be identical, i.e., $R_1 = R_2 = R$, and $C_1 = C_2 = C$, Equ.(19-5) reduces to

$$T = 1.386 RC \quad (19-6)$$

Therefore, the frequency of oscillation will be,

$$f = \frac{1}{0.69(C_1 R_3 + C_2 R_2)} = \frac{1}{1.38 RC}$$

19.4 Monostable Multivibrator

The circuit diagram of a monostable multivibrator is shown in Fig.19-16(a). As shown in Fig.8-16(b), the output of this multivibrator has a stable state and a quasi-stable state. The output goes to quasi-stable state when a trigger pulse is applied to the base of any transistor. The duration of the quasi-stable state will be: $T = 0.693R_T C_1$. After this time the output backs to the stable stage, and stays there until the next trigger pulse is applied.

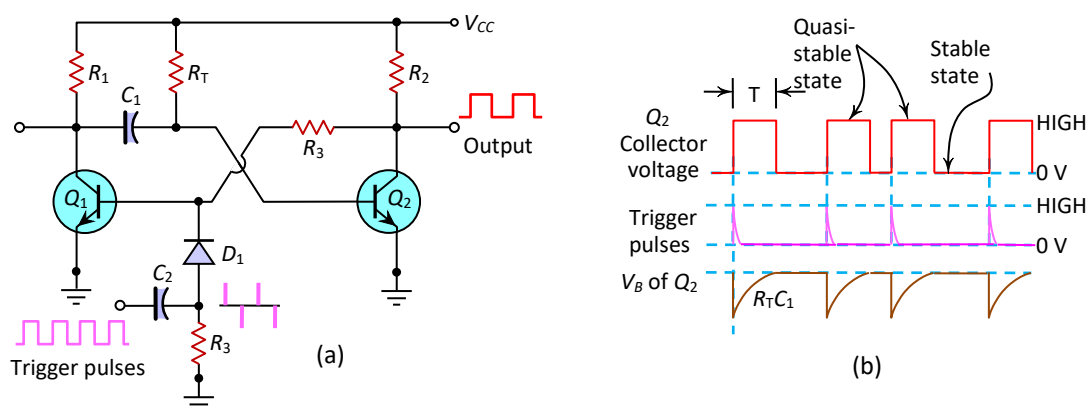


Fig.19-3: Monostable multivibrator: (a) Schematic diagram, and (b) Output waveforms

19.5 Bistable Multivibrator

A bistable multivibrator has 2 stable states. The output goes to HIGH state at one trigger pulse, and goes to LOW state at another trigger pulse and so on [as shown in Fig.19-4].

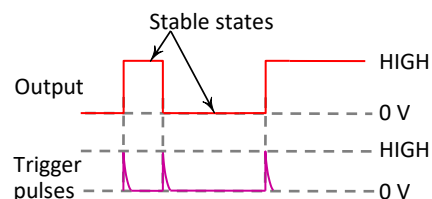
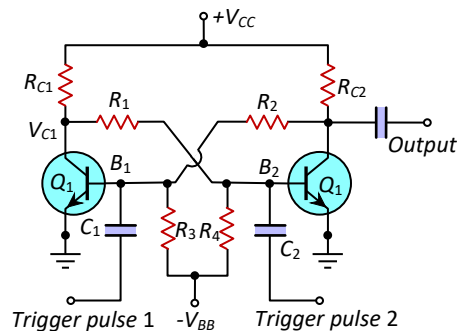


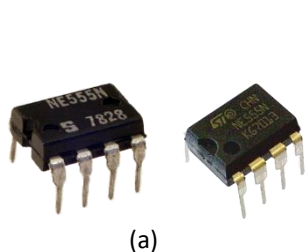
Fig.19-4: Output of bistable multivibrator

Circuit diagram of a bistable multivibrator



The Timer IC: 555

The **555 timer** is a commonly used IC, designed to produce a variety of output waveforms with the addition of an external RC network. Using this IC, different types of multivibrators can be designed.



(a)

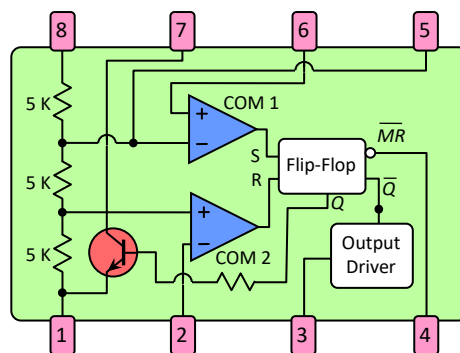


Fig.19-5: The 555 (timer) IC: (a) Photographs, and (b) Internal block diagram

1. Ground
2. Trigger
3. Output
4. Reset
5. Control voltage
6. Threshold
7. Discharge
8. Supply voltage (V_{CC})

19.6 Astable Multivibrator using 555 IC

The circuit diagram of an astable multivibrator using 555 timer is shown in Fig.19-6. The durations of HIGH and LOW times are controlled by two resistors R_1 , R_2 and one capacitor C_1 .

HIGH time: $t_1 = 0.69 (R_1 + R_2)C_1$

LOW time: $t_2 = 0.69 R_2 C_1$

Time period : $T = t_1 + t_2 = 0.69 (R_1 + R_2)C_1 + 0.69 R_2 C_1 = 0.69 (R_1 + 2R_2)C_1$.

Therefore, the frequency of oscillation will be,

$$f = \frac{1}{T} = \frac{1}{0.69(R_1 + 2R_2)C_1}$$

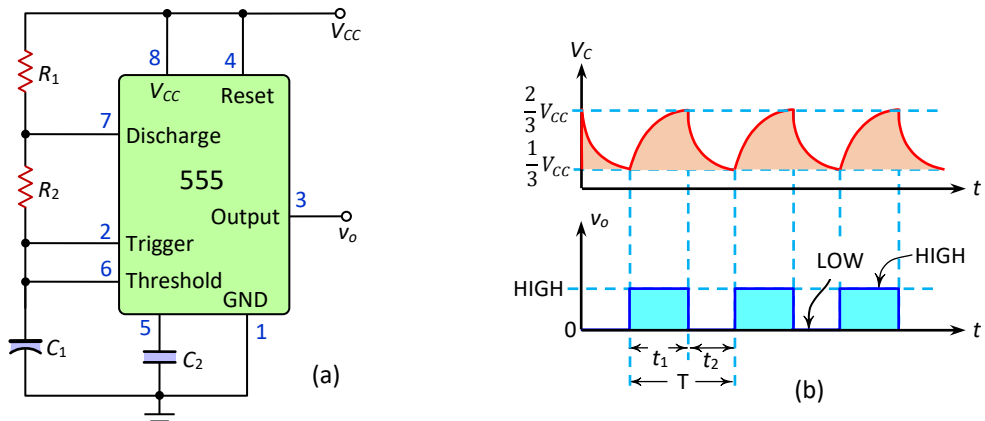
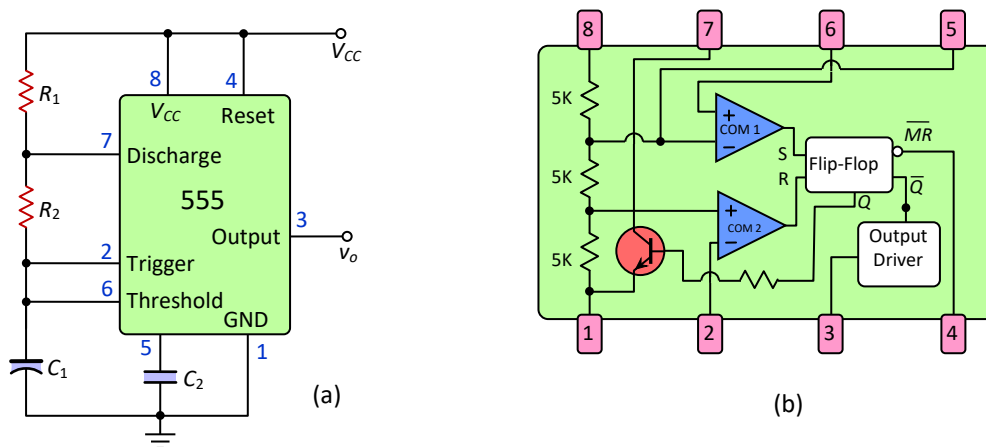
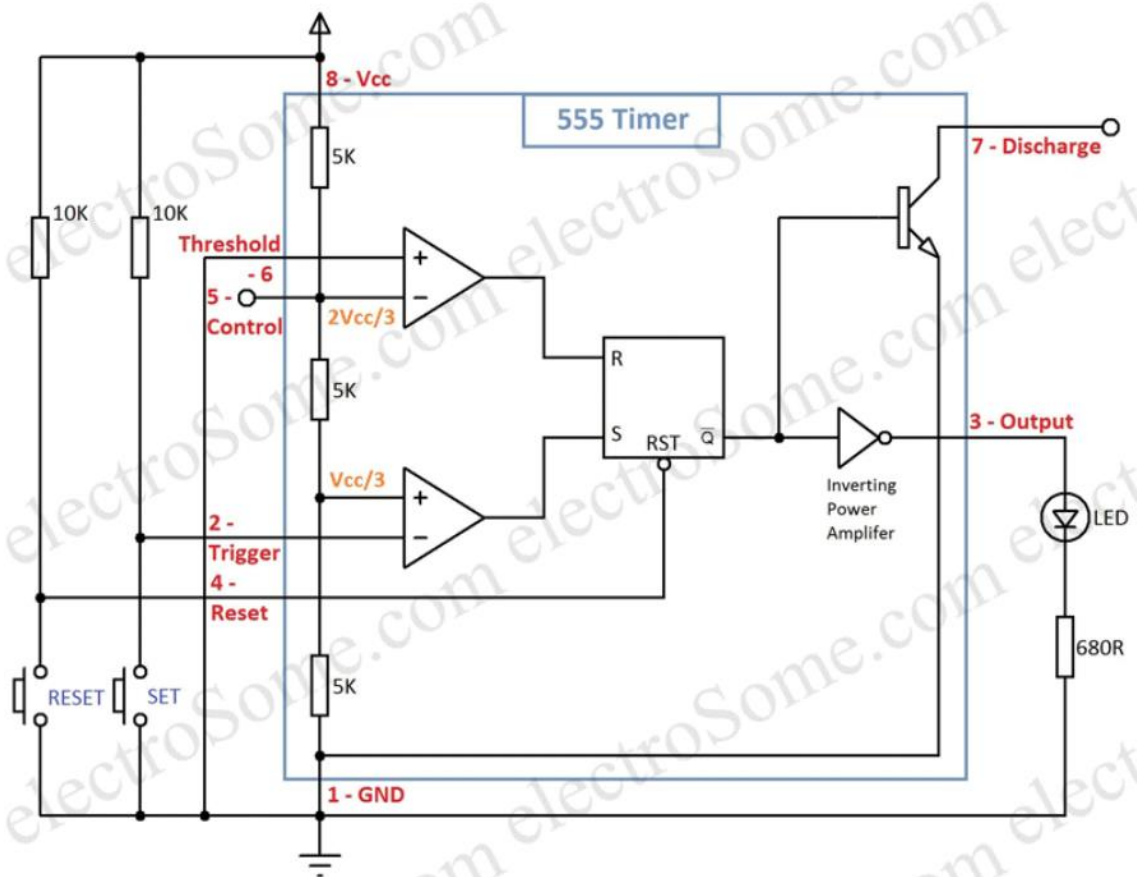


Fig.19-6: Astable multivibrator using 555 IC: (a) Schematic diagram, and (b) Output waveforms





19.7 Monostable Multivibrator

The circuit diagram of a monostable multivibrator using 555 timer is shown in Fig.19-7. The output of this multivibrator has a stable state and a quasi-stable state. As shown in Fig.19-7(b), the stable state is LOW and the quasi-stable state is HIGH. The output goes to the quasi-stable state when a trigger pulse is applied. After some time the output comes back to the stable stage, and stay there until the next trigger pulse is applied. Unlike the BJT monostable multivibrators, here a negative trigger pulse (less than $1/3$ of V_{CC}) is used. Monostable multivibrator is used to turn-ON a load (like a microwave oven) for a predefined amount of time and let it turn-OFF automatically.

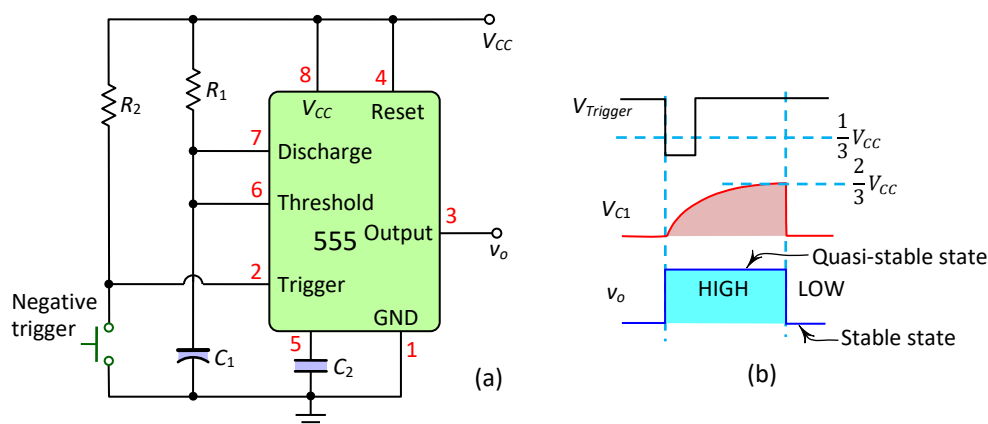


Fig.19-7: Monostable multivibrator using 555: (a) Schematic diagram, and (b) Output waveforms

Time duration of the quasi-stable state: $T = 1.1R_1C_1$

Astable Multivibrator Using Schmitt Trigger Circuit

Fig.19-8 shows how hysteresis can be introduced into a comparator. Here, the input is connected to the inverting terminal and reference voltage is applied to the noninverting terminal. Unlike the amplifiers, here feedback is applied to the noninverting terminal using a voltage divider network consisting of R_1 and R_2 which ensures positive feedback. Thus, this circuit is an inverting comparator circuit. Here, $UTL = LTL = R_1|V_{max}|/(R_1 + R_2) = \beta|V_{max}|$.

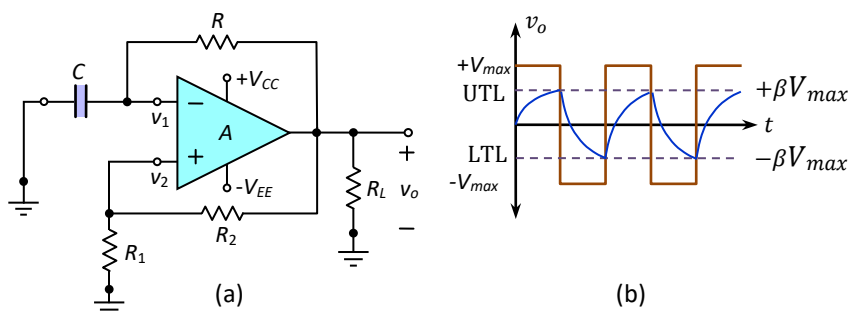


Fig.19-8 (a) Astable multivibrator using Schmitt trigger circuit, and (b) Output waveforms

A capacitor (C) is connected to the inverting input and a resistor (R) is connected to the inverting terminal and the output terminal. When the output is HIGH ($+V_{max}$), the capacitor will be charged through the resistor, R . For the HIGH ($=+V_{max}$) the voltage level at the non-inverting terminal will be $+\beta V_{max}$. When the capacitor voltage rises to $V_{max}\beta$, the op-amp will change its state and the output will be LOW ($= -V_{max}$). For the LOW ($= -V_{max}$) the voltage level at the non-inverting terminal will be $-V_{max}\beta$. For this negative output voltage the capacitor

will be discharged. When the capacitor voltage drops to $-V_{max}\beta$, the op-amp will change its state and the output will be HIGH again. In this way the output of the op-amp will continuously oscillates between HIGH and LOW states. The waveform of the output is shown in Fig. 19-8(b).

In this multivibrator HIGH time and LOW time will be equal, that is, the duty-cycle will be exactly 50%. Using Equ.(19-2), we can calculate the HIGH time and LOW time as,

$$t_H = t_L = RC \ln \left(\frac{V_t - V_i}{V_t - v_C} \right) = RC \ln \left(\frac{V_{max} + \beta V_{max}}{V_{max} - \beta V_{max}} \right)$$

or,

$$t_H = t_L = RC \ln \left(\frac{1 + \beta}{1 - \beta} \right)$$

Sweep Generator

VOLTAGE AND CURRENT SWEEP GENERATORS

Recent developments in electronics require extensive use of voltage and current sweep generators. We shall describe them here, starting with basic terminology.

TERMINOLOGY USED TO DESCRIBE SWEEP GENERATORS: Prior to discussing sweep circuits, we shall explain in brief the terminology used for such a description.

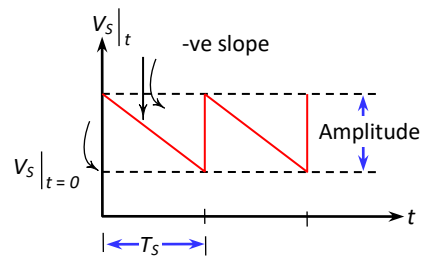
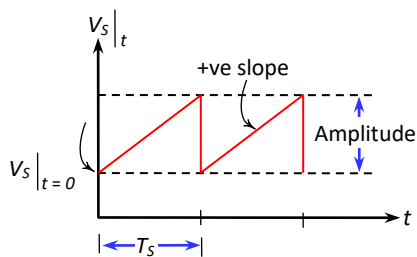
Ideal voltage ramp

In a linear ramp (also called sweep) generator, the voltage increases or de-creases linearly with time, and has duration. At the end of duration, it returns abruptly to initial value. In fig. 1 (a) ideal voltage sweep waveform (with positive slope) is shown. Voltage ramp (sweep) starts every time after a certain repetition period T_s , called ramp du-ration. Such an ideal ramp (sweep) waveform can be expressed as

$$or, \quad V_s|_t = V_s|_{t=0} + V \times t$$

where,

$V_s|_{t=0}$ is the initial value of the ramp voltage, and V is a constant that indicates the slope or speed (rate of change) for ideal voltage ramp waveform. In fig. 1 (b), ramp waveform having a negative slope is shown.



sl I V1 sr 1 t =0

Sweep onipPricie

757 --l '7i;9 Fig. 1. (a)

and

Sweep offipaude V I ' tt

Time Fig. 1. (b)

dV si it-0 dt

In fig. 2, practical sweep voltage waveform is shown. In it we note that voltage does not return abruptly to its initial value after the end of sweep duration, T_s . Thus it takes some more time after the end of sweep duration, T_s , to come to its initial value. This is called retrace interval or flyback period or restoration time period, and is denoted by T_r . It is

tcrs VI t=0

Sweep Retrace

7;z Fig. 2.

Time

560

Sweep Retrace Reset

.4—'1 Time $T_{s,r}$, 7,,,reset. Fig. 3.

